Name:

BC Test #3 Review

KEY

1) The Maclaurin series for a function f is given by $\sum_{n=1}^{\infty} \frac{(n+1)x^n}{n^2 6^n}$ and converges to f(x) for all x in the interval of convergence. It can be shown that the Maclaurin series for f has a radius of convergence of 6.

a) Determine whether the Maclaurin series for f converges or diverges at $x = 6$. Give a reason for your answer.

b) Find the general term of the Maclaurin series for f ', the derivative of f. Find the radius of convergence of the Maclaurin series for f'.

c) Let $g(x) = \sum_{n=1}^{\infty} \frac{(n+1)x^{2n}}{n^2 3^n}$. Use the ratio test to determine the radius of convergence of the Maclaurin series for g.

a) $\sum_{n=1}^{\infty} \frac{(n+1)\cancel{6^n}}{n^2 \cancel{6^n}} \to \sum_{n=1}^{\infty} \frac{n+1}{n^2}$

Compare $\sum_{n=1}^{\infty} \frac{1}{n}$ p-series $p=1$, $0<p\le 1$

The series diverges

$\lim_{n\to\infty} \frac{n+1}{n^2} \cdot \frac{n}{1} = \lim_{n\to\infty} \frac{n^2+n}{n^2} = 1 > 0$ and finite

Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, $\sum_{n=1}^{\infty} \frac{n+1}{n^2}$ also diverges by LCT

b) general term of f: $\frac{(n+1)x^n}{n^2 6^n}$ general term of f': $\frac{(n+1)\cancel{n}x^{n-1}}{n^{\cancel{2}} 6^n} = \frac{(n+1)x^{n-1}}{n\,6^n}$

RTAC $\lim_{n\to\infty} \left| \frac{(n+1+1)x^{n+1-1}}{(n+1)6^{n+1}} \cdot \frac{n6^n}{(n+1)x^{n-1}} \right| = \lim_{n\to\infty} \left| \frac{(n+2)x^n}{(n+1)6^{n+1}} \cdot \frac{n6^n}{(n+1)x^{n-1}} \right|$

$= \lim_{n\to\infty} \left| \frac{(n+2)\cancel{x^n}}{(n+1)\cancel{6^n}\cdot 6^1} \cdot \frac{n\cancel{6^n}}{(n+1)\cancel{x^n}\cdot x^{-1}} \right| = \lim_{n\to\infty} \left| \frac{n(n+2)}{6(n+1)^2} x \right| = \left|\frac{x}{6}\right| < 1$ $\boxed{\text{ROC: } 6}$

$\hookrightarrow -6 < x < 6$

c) RTAC $\lim_{n\to\infty} \left| \frac{(n+1+1)x^{2(n+1)}}{(n+1)^2 3^{n+1}} \cdot \frac{n^2 3^n}{(n+1)x^{2n}} \right| = \lim_{n\to\infty} \left| \frac{(n+2)x^{2n+2}}{(n+1)^2 3^{n+1}} \cdot \frac{n^2 3^n}{(n+1)x^{2n}} \right|$

$= \lim_{n\to\infty} \left| \frac{(n+2)\cancel{x^{2n}}\cdot x^2}{(n+1)^2 \cancel{3^n}\cdot 3^1} \cdot \frac{n^2 \cancel{3^n}}{(n+1)\cancel{x^{2n}}} \right| = \lim_{n\to\infty} \left| \frac{n^2(n+2)x^2}{3(n+1)^3} \right| = \left|\frac{x^2}{3}\right| < 1$

$\hookrightarrow -1 < \frac{x}{3} < 1$

$\hookrightarrow -3 < x < 3$

$\boxed{\text{ROC: } 3}$

2) The Taylor series for a function f about x=1 is given by $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n}{n}(x-1)^n$ and converges to f(x) for |x-1|<R, where R is the radius of convergence of the Taylor series.

a) Find the value of R.

b) Find the first three nonzero and the general term of the Taylor series for f ', the derivative of f, about x=1.

c) The Taylor series for f ' about x=1, found in part (b), is a geometric series. Find the function f ' to which the series converges for |x-1|<R. Use this function to determine f for |x-1|<R.

a) RTAC $\lim_{n\to\infty}\left|\frac{2^{n+1}(x-1)^{n+1}}{(n+1)}\cdot\frac{n}{2^n(x+1)^n}\right| = \lim_{n\to\infty}\left|\frac{\cancel{2^n}\cdot 2^1\cancel{(x-1)^n}\cdot(x-1)^1}{(n+1)}\cdot\frac{n}{\cancel{2^n}\cancel{(x+1)^n}}\right|$

$= \lim_{n\to\infty}\left|\frac{2n(x-1)}{(n+1)}\right| = |2(x-1)|<1 \rightarrow -\frac{1}{2}<\frac{2(x-1)}{2}<\frac{1}{2} \rightarrow -\frac{1}{2}<(x-1)<\frac{1}{2}$

$\boxed{R=1/2}$

b) $f(x)=2(x-1)-\frac{2^2}{2}(x-1)^2+\frac{2^3}{3}(x-1)^3+\dots+(-1)^{n+1}\frac{2^n}{n}(x-1)^n+\dots$

$\boxed{f'(x)\ 2-2^2(x-1)+2^3(x-1)^2+\dots+(-1)^{n+1}2^n(x-1)^{n-1}+\dots}$

c) $f'(x)=2-\overbrace{2^2(x-1)}^{-2(x-1)}+2^3(x-1)^2+\dots+(-1)^{n+1}2^n(x-1)^{n-1}+\dots$

$f'(x)=\frac{2 \leftarrow a}{1-(-2(x-1))}$ (geometric sum)

$\uparrow r$

$f'(x)=\frac{2}{1-(-2x+2)}=\frac{2}{1+2x-2}=\frac{2}{2x-1}, \quad -\frac{1}{2}<(x-1)<\frac{1}{2}$

$f'(x)=\frac{2}{2x-1} \rightarrow f(x)=\int\frac{2}{2x-1}dx = \ln|2x-1|+C$

$f(1)=2(x-1)$
$=2(1-1)$
$f(1)=0$

$0=\ln|2(1)-1|+C$
$0=\ln|2-1|+C$
$0=\ln|1|+C$
$0=0+C$
$C=0$

$\rightarrow \boxed{f(x)=\ln|2x-1|, \ -\frac{1}{2}<(x-1)<1/2}$

3) $f(0) = 0$
$f'(0) = 1$
$f^{n+1}(0) = -n \cdot f^{(n)}(0)$ *for all* $n \geq 1$

A function f has derivatives of all orders for $-1 < x < 1$. The derivatives of f satisfy the conditions above. The Maclaurin series for f converges to f(x) for |x|<1.

a) Show that the first four nonzero terms of the Maclaurin series for f are $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$, and write the general term of the Maclaurin series for f.

b) Determine whether the Maclaurin series described in part a) converges absolutely, converges conditionally, or diverges at x=1. Explain your reasoning.

c) Write the first four nonzero terms and the general term of the Maclaurin series for $g(x) = \int_0^x f(t)dt$

a) $f(0)=0$, $f'(0)=1$, $\underset{n=1}{f''(0)} = -1 \cdot f'(0) = -1(1) = -1$, $\underset{n=2}{f'''(0)} = -2 \cdot f''(0) = -2(-1) = 2$

$\underset{n=3}{f^4(0)} = -3 \cdot f'''(0) = -3(2) = -6$

$$\boxed{1x + \frac{(-1)x^2}{2!} + \frac{2}{3!}x^3 + \frac{(-6)}{4!}x^4} \quad \boxed{\text{general}: \frac{(-1)^{n+1}x^n}{n}}$$

b) plug in x=1 to general: $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}(1)^n}{n} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{n}$

- $\left|\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{n}\right| = \sum_{n=0}^{\infty} \frac{1}{n}$ p-series $p=1$, $0<p\leq 1$
  The series diverges

- $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{n}$ $a_n = \frac{1}{n}$ $a_{n+1} = \frac{1}{n+1}$ $a_n > a_{n+1}$ $\lim_{n\to\infty} \frac{1}{n} = 0$ The series converges by AST.

Therefore the series converges conditionally.

c) $\int_0^x \left(t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots + \frac{(-1)^{n+1}t^n}{n}\right) dt$

$$= \boxed{\frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{12} - \frac{x^5}{20} + \dots + \frac{(-1)^{n+1}x^{n+1}}{n(n+1)}}$$

4) The function f has a Taylor series about x=1 that converges to f(x) for all x in the interval of convergence. It is known that f(1)=1, f '(1)= -½, and the nth derivative of f at x=1 is given by $f^n(1) = (-1)^n \frac{(n-1)!}{2^n}$ for $n \geq 2$.

a) Write the first four nonzero terms and the general term of the Taylor aries for f about x=1.

b) The Taylor series for f about x=1 has a radius of convergence of 2. Find the interval of convergence. Show the work that leads to your answer.

c) The Taylor series for f about x=1 can be used to represent f(1.2) as an alternating series. Use the first three nonzero terms of the alternating series to approximate f(1.2).

a) $f(1)=1, f'(1)=-\frac{1}{2}, \underset{n=2}{f''(1)} = \frac{(-1)^2(2-1)!}{2^2} = f''(1) = \frac{1}{4}, \underset{n=3}{f'''(1)} = \frac{(-1)^3(3-1)!}{2^3} = \frac{-2}{8} = -\frac{1}{4}$

$\rightarrow 1 + (\frac{-1}{2})(x-1) + \frac{(1/4)(x-1)^2}{2!} + \frac{(-1/4)(x-1)^3}{3!} + \dots + \frac{\frac{(-1)^n(x-1)^n}{2^n}}{n!}(x-1)^n + \dots$

$\rightarrow \boxed{1 - \frac{1}{2}(x-1) + \frac{1}{8}(x-1)^2 - \frac{1}{24}(x-1)^3 + \dots + \frac{(-1)^n(x-1)^n}{2^n \cdot n} + \dots}$

b) Center=1, ROC:2 $\rightarrow -1 < x < 3$ (Test endpoints for $1 + \sum_{n=1}^{\infty} \frac{(-1)^n(x-1)^n}{2^n \cdot n}$)

when x=-1:

$1 + \sum_{n=1}^{\infty} \frac{(-1)^n(-2)^n}{2^n n} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n(-1)^n \cancel{2^n}}{\cancel{2^n} \cdot n} = 1 + \sum_{n=1}^{\infty} \frac{1}{n}$ p-series, $0<p\leq 1$, $p=1$ The series diverges

when x=3:

$1 + \sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{2^n n} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ $a_n = \frac{1}{n}$ $a_{n+1} = \frac{1}{n+1}$, $a_n > a_{n+1}$ $\lim_{n\to\infty} \frac{1}{n} = 0$, The series converges by AST.

$\boxed{\text{IOC: } (-1, 3]}$

c) $1 - \frac{1}{2}(x-1) + \frac{1}{8}(x-1)^2$

$\boxed{f(1.2) \approx 1 - \frac{1}{2}(1.2-1) + \frac{1}{8}(1.2-1)^2}$